

A Double Fittings Based Total Variation Model for Image Denoising

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Abstract—In this paper, weighted image gradients based fitting is introduced into the proposed model for image denoising, which is called double fittings based total variation (DFTV) model. The weighted gradients are designed to maintain image structures during the denoising process. The definition of the weight function makes it better to enhance edges of images and maintain structures. For solving the proposed model, alternating direction method of multipliers (ADMM) is explored. In order to show the effectiveness and efficiency of the proposed model, both of the visual results and quantity evaluation are provided for compared experiments.

Keywords—Image denoising, total variation, weighted gradients, ADMM

I. INTRODUCTION

Image denoising is a widely explored yet challenging problem in image processing field. In practice, images are often disturbed by noise, which arises quality degradation of images.

It is known that restoring a clean image from a noisy one is ill-posed [1], which requires mathematical theories and methods. Many models and techniques were proposed for denoising, such as total variation model [2], partial differential equation model [3], machine learning techniques [4,5] etc. Among them, the total variation based models are one of the most popular categories. They are applied to many image problems, such as image segmentation [6,7], image deblurring [8,9], image decomposition [10], etc. Many denosing models are based on the well known total variation (TV) model [11] (also called ROF model) and try to improve the staircase effect [12] of TV. In [13], a high order total variation (HOTV) based model was proposed yet resulted in blur edges. Some models used different orders of differentiation for denoising, such as fractional variable order differential defined in [14], TGV [15], etc. Regularization is very important to modeling. For example, in [16] $l_p - \alpha l_q$ based regularization is explored, which is a combination of different norms. Graph laplacian-based regularization in [17] is also very effective to denoising. Most models mentioned above are focused on regularization, which are to approximately describe the sparsity of images. It can be seen that image properties are better described by suitable regularization and textures are well kept in denoising processing. However, the fidelity term is not

supposed to be neglected, which is critical to maintaining image structures. In this sense, a novel fidelity term is introduced into the proposed model in this paper. The details are seen in Section 3.

For computing of TV based models, alternating direction method of multipliers (ADMM) [18] is widely explored due to its highly efficiency and brief programming. Therefore in this paper we use ADMM to solve the proposed model.

The rest of the paper is organized as follows. In Section 2, some notations and related models are introduced briefly. In Section 3, the proposed model is described in details and its calculation is discussed. Section 4 provides compared experiments to show the effectiveness and efficiency of the proposed model. Section 5 concludes this paper.

II. PRELIMINARIES

In this section, some notations and related models are introduced.

A. Notations

Given noisy image $f: \Omega \rightarrow R$, it is represented by a function from image region Ω to real number set R . The size of f is denoted by $M \times N$. The gradient of f is $\nabla f = (\partial_x f, \partial_y f)^T$, $\partial_x f$ and $\partial_y f$ are gradients of f in x and y directions respectively. They are calculated by

$$\partial_x f_{i,j} = \begin{cases} f_{i,j} - f_{i-1,j}, & 1 < i \leq M, 1 \leq j \leq N \\ f_{i,j} - f_{M,j}, & i = 1, 1 \leq j \leq N \end{cases} \quad (1)$$

$$\partial_y f_{i,j} = \begin{cases} f_{i,j} - f_{i,j-1}, & 1 \leq i \leq M, 1 < j \leq N \\ f_{i,j} - f_{i,N}, & 1 \leq i \leq M, j = 1 \end{cases} \quad (2)$$

∂_x^* and ∂_y^* are adjoint operators of ∂_x and ∂_y respectively. They are computed by

$$\partial_x^* f_{i,j} = -\partial_x f_{i,j}, \quad \partial_y^* f_{i,j} = -\partial_y f_{i,j} \quad (3)$$

This work is supported by the National Natural Science Foundation of China (No. 61671063).

B. TV model

TV model was proposed by Rudin et al in [11] and its discretized form is

$$\min_u \frac{\lambda}{2} \|f - u\|_2^2 + \|\nabla u\|_1 \quad (4)$$

In this model, l_1 norm of image gradient is used for regularization. It performs well in maintaining edges during denoising. This model is widely applied to many image problems even though it may generate staircase effect.

C. HOTV model

The HOTV [13] model utilizes high order differential of image gradients to regularization and formulated as

$$\min_u \frac{\lambda}{2} \|f - u\|_2^2 + \|\nabla^2 u\|_1 \quad (5)$$

It was proposed to improve the staircase of TV model. However, it may result in blur edges and high order differential takes more computational time.

D. ATV model

Recently, Z. Pang et al proposed a new anisotropic total-variation-based model (ATV) [19] to remove noises. The ATV model is described as

$$\begin{aligned} \min_{u,v,w} \frac{\lambda}{2} \|u - f\|_2^2 + \|w\|_{2,1} \\ \text{s.t. } w = Tv, v = \nabla u \end{aligned} \quad (6)$$

In this model, the authors consider anisotropic property and incorporate a weight function T into the regularization, which makes it better to describe local features of images though two extra parameters are added in their weight function. The ATV model pays more attention to local image features and performs well in image denoising.

The models above consider more about the regularization for maintaining details during denoising. However, structure is very important information of an image. Therefore, in the basis of the models mentioned before, structure information is explored and incorporated into the fitting terms of our proposed denoising model.

III. THE PROPOSED MODEL

The proposed double fittings based total variation (DFTV) model for zero mean Gaussian noise is formulated as

$$\min_u \frac{\lambda}{2} \|f - u\|_2^2 + \alpha \|h - \nabla u\|_2^2 + \|\nabla u\|_1 \quad (7)$$

where h is the weighted gradients of noisy image f , i.e.,

$$h = w \cdot \nabla f, \quad w = \left(\exp(|\partial_x f|), \exp(|\partial_y f|) \right). \quad (8)$$

As follows, in order to make it more clear, we first give the explanation of motivation for modeling and then discuss the computation of our model.

A. Motivation

Among many denoising models, one of the most popular fittings is $\|f - u\|^2$, where f is an image polluted by Gaussian noise and u is the desired image. Besides observed images themselves, gradient is a very important feature of image structure. Therefore, it is reasonable and necessary to incorporate structure information to models. Taking image Lena as an example, in Fig. 1, (a) and (b) are gradient amplitudes of clean and noisy images respectively while (c) is the weighted gradient amplitudes of noisy image. The noisy images are with zero means Gaussian noise. It is seen that for image f with Gaussian noise, lower frequencies are more vulnerable to turbulence than higher ones. With the definition of weight function w , $w\nabla f$ is helpful to maintain lower frequencies and enhance higher frequencies. In this sense, $w\nabla f$ is close to gradients of the clean image and hence is more suitable to be the complement of fitting. Therefore, weighted gradients based fitting is introduced in the proposed DFTV model.

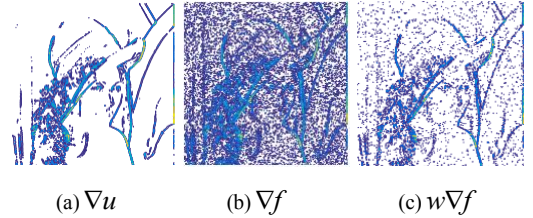


Fig.1. Gradients of Lena under different situations.

B. Algorithm

For convenience, two variables u_1 and u_2 are introduced, where $u_1 = \partial_x u$ and $u_2 = \partial_y u$. Then the model (7) can be formulated as the following optimization problem

$$\begin{aligned} \min_{u, u_1, u_2} \frac{\lambda}{2} \|f - u\|_2^2 + \frac{\alpha}{2} \|h - \nabla u\|_2^2 + \|(u_1, u_2)\|_1 \\ \text{s.t. } u_1 = \partial_x u, u_2 = \partial_y u \end{aligned} \quad (9)$$

For solving model (9), the efficient ADMM is explored.

The Lagrangian of (9) is formulated as

$$\begin{aligned} L(u, u_1, u_2; \gamma_1, \gamma_2) \\ = \frac{\lambda}{2} \|f - u\|_2^2 + \frac{\alpha}{2} \|h - \nabla u\|_2^2 + \|(u_1, u_2)\|_1 + \langle \gamma_1, u_1 - \partial_x u \rangle \\ + \langle \gamma_2, u_2 - \partial_y u \rangle + \frac{\tau}{2} \left(\|u_1 - \partial_x u\|_2^2 + \|u_2 - \partial_y u\|_2^2 \right). \end{aligned} \quad (10)$$

Solving model (9) is equivalent to the following problem

$$\max_{\gamma_1, \gamma_2} \min_{u, u_1, u_2} L(u, u_1, u_2; \gamma_1, \gamma_2). \quad (11)$$

In order to solve problem (11), we separate it into several sub-problems as follows.

$$u^* = \arg \min_u \left\{ \frac{\lambda}{2} \|f - u\|^2 + \frac{\alpha}{2} \|h - \nabla u\|^2 + \frac{\tau}{2} \left(\|u_1 - \partial_x u + \gamma_1/\tau\|^2 + \|u_2 - \partial_y u + \gamma_2/\tau\|^2 \right) \right\} \quad (12)$$

$$u_1^* = \arg \min_{u_1} \{ \|u_1 - \partial_x u + \gamma_1/\tau\|^2 + \|u_1\|_1 \} \quad (13)$$

$$u_2^* = \arg \min_u \left\{ \frac{\tau}{2} \|u_2 - \partial_y u + \gamma_2/\tau\|^2 + \|u_2\|_1 \right\} \quad (14)$$

Subproblems (12), (13) and (14) are to be solved successively. Minimizing (12) with respect to u , we can obtain the following formula

$$\begin{aligned} (\lambda I + (\alpha + \tau) \nabla^* \nabla) u &= \lambda f + \alpha \nabla^* h + \tau \partial_x^* (u_1 + \gamma_1/\tau) \\ &\quad + \tau \partial_y^* (u_2 + \gamma_2/\tau). \end{aligned} \quad (15)$$

With periodic boundary condition and forward difference, the left side of (15) are circulant matrices which can be dia-

gonalized by fast Fourier Transform [20]. Subproblem (12) can be efficiently computed by

$$u = F^{-1} \left[\frac{\lambda F(f + \alpha \nabla^* h) + \tau F(\partial_x^* (u_1 + \gamma_1/\tau) + \partial_y^* (u_2 + \gamma_2/\tau))}{F(\lambda I) + (\alpha + \tau) F(\nabla^* \nabla)} \right]. \quad (16)$$

$$u_1 = \max \left(1 - \frac{1}{\tau \cdot s}, 0 \right) \cdot s_1 \quad (17)$$

$$u_2 = \max \left(1 - \frac{1}{\tau \cdot s}, 0 \right) \cdot s_2 \quad (18)$$

where $s_1 = \partial_x u - \gamma_1/\tau$, $s_2 = \partial_y u - \gamma_2/\tau$, $s = \sqrt{s_1^2 + s_2^2}$.

The computation steps are summarized in Algorithm 1. In Algorithm 1, *MaxIter* is the max number of iterations, *tol* is the tolerance. The termination criterion is defined by the relative error between the k -th and $(k+1)$ -th iterations as

$$err = \frac{\|u^{k+1} - u^k\|}{\|u^k\|} \leq tol. \quad (19)$$

TABLE I. COMPARISON OF PSNR AND SSIM WITH GAUSSIAN NOISE VARIANCES $\sigma = 0.05$ AND 0.1

σ	Models	<i>Lena</i>		<i>Panda</i>		<i>Pepper</i>		<i>Bird</i>		<i>Man</i>		<i>Baboon</i>	
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
0.05	TV	28.1641	0.8439	28.4822	0.7598	28.5167	0.8766	23.7697	0.7673	23.5203	0.7852	22.1085	0.6287
	HOTV	30.2342	0.8698	30.3109	0.7856	30.9611	0.8988	27.8503	0.8634	28.9850	0.8650	27.8941	0.8477
	ATV	31.4223	0.8595	31.0329	0.8061	31.0800	0.8985	29.7445	0.8766	30.1060	0.8829	28.2097	0.8506
	DFTV	31.5205	0.8856	31.1256	0.8097	32.2509	0.9026	29.5260	0.8807	30.3305	0.8855	28.2365	0.8501
0.1	TV	27.3333	0.7915	27.6613	0.7057	27.6265	0.8158	23.4908	0.6875	23.5021	0.7171	21.9757	0.5562
	HOTV	27.2440	0.7955	27.4884	0.7007	27.7085	0.8332	24.5106	0.7319	25.9414	0.7752	23.4003	0.6385
	ATV	28.0257	0.7788	27.8814	0.6938	28.5321	0.8115	25.7621	0.7251	26.8203	0.7434	24.0255	0.6709
	DFTV	28.1274	0.8045	28.2010	0.7169	28.6719	0.8346	25.5842	0.7594	26.8214	0.8084	24.0648	0.6839

Algorithm 1: ADMM for solving (9)

Initialize: $\gamma_1, \gamma_2, u_1, u_2$

Input: $\lambda, \tau, \text{MaxIter}$

Output: u

For $k=1:\text{MaxIter}$

Update: u^{k+1} by Eq. (16)

u_1^{k+1}, u_2^{k+1} by Eq. (17)-(18)

$\gamma_1^{k+1} \leftarrow \gamma_1^k + \tau(u_1^k - \partial_x u^{k+1})$

$\gamma_2^{k+1} \leftarrow \gamma_2^k + \tau(u_2^k - \partial_y u^{k+1})$

If (19) is satisfied

break

end

End

IV. EXPERIMENTS

In this section, compared experiments of TV, HOTV, ATV and the proposed DFTV models are conducted. All the numerical experiments are performed via MATLAB (R2016a) on a windows 7 (64bit) desktop computer with an Intel Core i5 3.20 GHz processor and 4.0 GB of RAM. All the test images for denoising experiments are scaled to $[0,1]$. For quality evaluation, two important indicators, peak-signal-to-noise ration (PSNR) and the structural similarity index (SSIM) [22] are compared. PSNR is to measure the denoising effect and SSIM reflects the similarity between noisy and denoised images. The larger values of PSNR and SSIM means better denoising results. They are computed by MATLAB command *psnr* and respectively.

In experiments, parameters of the proposed DFTV model λ, α are tunable, $\tau \in (0,1)$. Seen in Fig. 2, the four columns from left to right are results of TV, HOTV, ATV and the proposed DFTV models in turn. They are denoised from observed images polluted by Gaussian noise with variance $\sigma = 0.05$. In Fig. 3, they are compared results in case of $\sigma = 0.1$. Seen from the two figures, the proposed DFTV model obtains more desirable images. Especially for image Man (the fifth rows in Fig. 2 and Fig. 3), the effect is more obvious. Apart from visual effects, quantity evaluation is also provided. In order to show the effectiveness of the proposed DFTV model, PSNR and SSIM are compared in Table I, where the Gaussian noise variances are 0.05 and 0.1 respectively. It is seen that the proposed DFTV model results in higher PSNR and SSIM values, which shows the effectiveness of the proposed DFTV model.

TABLE II. COMPARISON OF COMPUTATIONAL TIME (UNIT: SECOND).

σ	Models	Lena	Panda	Pepper	Bird	Man	Baboon
0.05	TV	4.46	4.45	4.40	4.66	0.91	25.50
	HOTV	12.61	12.73	12.11	12.63	2.50	40.92
	ATV	2.61	2.55	2.94	3.74	0.86	8.17
	DFTV	1.68	1.49	1.60	1.96	0.50	3.88
0.1	TV	4.62	4.50	4.55	4.56	0.89	25.61
	HOTV	12.13	12.12	12.60	11.79	2.31	68.60
	ATV	3.43	3.25	3.53	4.02	0.77	12.82
	DFTV	2.22	2.67	2.22	2.71	0.60	4.64

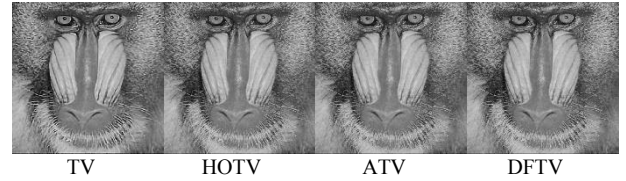


Fig. 2. Compared results for images with $\sigma = 0.05$.

For efficiency, computational time is provided in Table II for different noise levels. It shows that the ATV model and the proposed DFTV model cost less computational time than TV and HOTV models. Different from TV and HOTV models, DFTV and ATV models are with weight functions in modeling while the weight function of the proposed DFTV is more efficient than that of ATV model.

V. CONCLUSION

In this paper, in order to better maintain images structure during the denoising process, weighted image gradients based fitting is designed and introduced to the proposed DFTV model. The definition of the weight function is defined according to the given images without any extra parameters, which is efficient to enhance higher frequencies such as edges. This design benefits of maintaining image structures during the denoising processing and is convenient for programming.



Fig. 3. Compared results for images with $\sigma = 0.1$.

Compared with TV, HOTV and ATV models, the proposed DFTV model achieves larger values of PSNR and SSIM, and shorter computational time. The results demonstrates the effectiveness and efficiency of the proposed model.

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